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1 Continuity at a point

Previously, we have said that a function is continuous at a point a if its graph does not have a hole or break at a; that is, the graph near a can be drawn without lifting your pencil. As you might have come to expect at this point, this intuitive but vague definition is insufficient for all but the simplest functions.

Example 1.1. Consider the function

$$h(x) = \begin{cases} x\sin\frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Check out its graph `https://www.desmos.com/calculator/xi5qbi8har` and see if you think it is continuous at 0.

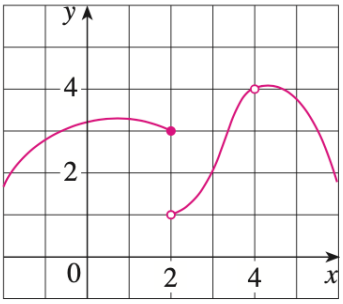
Intuitively, a function which is continuous at a point a should map inputs near a to outputs near $f(a)$. This sounds a lot like a limit! To help us formulate a precise definition of continuity, let's use the graph of the function f below to find the following:

Exercise 1.2. Use the graph of the function f below to find the following:

(a) $\lim_{x\to 2^-} f(x)$ (b) $\lim_{x\to 2^+} f(x)$ (c) $\lim_{x\to 2} f(x)$ (d) $f(2)$

(e) $\lim_{x\to 4^-} f(x)$ (f) $\lim_{x\to 4^+} f(x)$ (g) $\lim_{x\to 4} f(x)$ (h) $f(4)$

(i) $\lim_{x\to 3^-} f(x)$ (j) $\lim_{x\to 3^+} f(x)$ (k) $\lim_{x\to 3} f(x)$ (l) $f(3)$



Using your answers above, what condition(s) must hold for a function f to be continuous at a point a?

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Definition 1.3 (Continuity at a point)**.** A function f is *continuous at a point* a if $\lim_{x\to a} f(x) = f(a)$. If f is not continuous at a point a, then we say f is *discontinuous* at a.

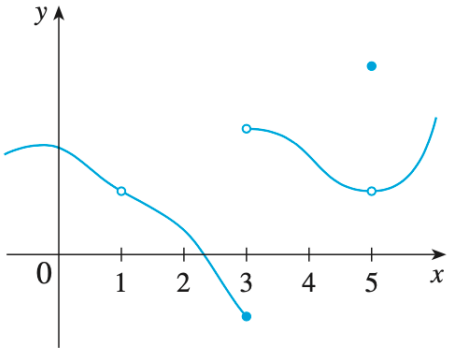
Note that if f is continuous at a, then three independent things must be true:

(i) $f(a)$ is defined

(ii) $\lim_{x\to a} f(x)$ exists

(iii) these are equal

Exercise 1.4. Use the graph of the function f below to list all points where f is discontinuous. At each point of discontinuity, state why f is discontinuous there by using the checklist above (i.e., state which of the conditions (i)-(iii) fails).

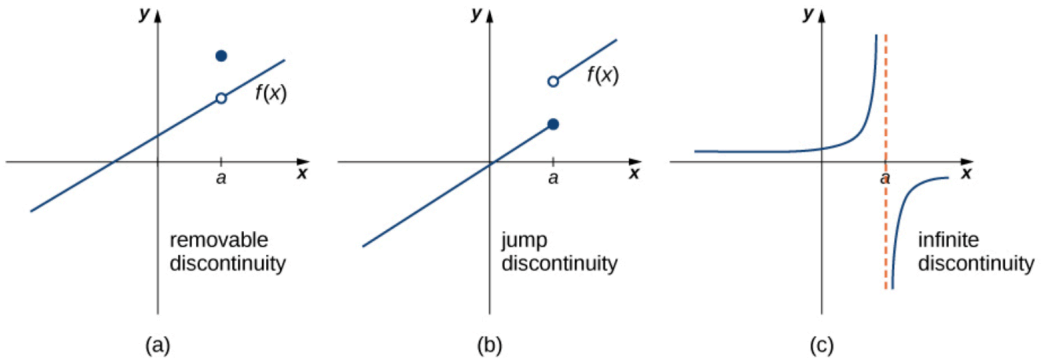


Definition 1.5 (Classification of discontinuities)**.** If $f(x)$ is discontinuous at a, then one says that

(a) f has a *removable discontinuity* if $\lim_{x\to a} f(x) = L \in \mathbb{R}$ (in this case either $f(a)$ is undefined, or $f(a)$ is defined by $L \neq f(a)$).

(b) f has a *jump discontinuity* if $\lim_{x\to a^-} f(x) = L_1 \in \mathbb{R}$ and $\lim_{x\to a^+} f(x) = L_2 \in \mathbb{R}$ but $L_1 \neq L_2$.

(c) f has an *infinite discontinuity* at a if $\lim_{x\to a^-} f(x) = \pm\infty$ or $\lim_{x\to a^+} f(x) = \pm\infty$



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Remark 1.6. Though we will rarely encounter this situation, if either $\lim_{x\to a^-} f(x)$ or $\lim_{x\to a^+} f(x)$ fails to exist (not even as $\pm\infty$), then f is said to have an *essential discontinuity* at a. For example, $\cos(\frac{1}{x})$ has an essential discontinuity at 0 since neither one sided limit exists due to the way $\cos(\frac{1}{x})$ oscillates between 1 and -1 faster and faster as $x \to 0$ from either side.

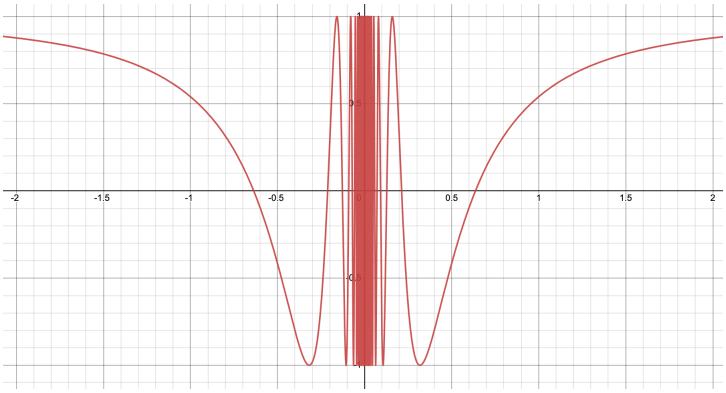


Figure 1: Graph of $\cos(\frac{1}{x})$.

(You may skip the following proof on a first reading and go directly to the exercises at the beginning of the next page.) To prove that $\lim_{x\to a^+} f(x)$ does not exist, consider points of the form $x_n = \frac{1}{n\pi}$, where $n \in \mathbb{N}$. Then

$$\cos\left(\frac{1}{x_n}\right) = \cos(n\pi) = \begin{cases} 1 & \text{if } n \text{ is even,} \\ -1 & \text{if } n \text{ is odd.} \end{cases}$$

Let $0 < \varepsilon < 2$. For any $\delta > 0$, there exists $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < \delta$. Since $\pi > 1$ and $n > 0$,

$$n\pi > n > 0.$$

Since $n+1 > n$ and $\pi > 0$,

$$(n+1)\pi > n\pi.$$

Putting these together gives

$$(n+1)\pi > n\pi > n > 0$$

and therefore

$$0 < \frac{1}{(n+1)\pi} < \frac{1}{n\pi} < \frac{1}{n} < \delta.$$

This shows that, for every $\delta > 0$, the open interval $(0, \delta)$ contains x_n and x_{n+1} for some $n \in \mathbb{N}$, and therefore the function $\cos(\frac{1}{x})$ attains the values $+1$ and -1 on $(0, \delta)$. Since $|\cos(x_n) - \cos(x_{n+1})| = 2 > \varepsilon$, $\lim_{x\to 0^+} \cos(\frac{1}{x})$ does not exist.

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2 Exercises

Determine whether the given function is continuous at the given point. If it is not continuous, classify the type of discontinuity.

1. $f(x) = \dfrac{x^2-4}{x-2}$ at $x = 2$.

2. $f(x) = \begin{cases} -x^2+4 & \text{if } x \leq 3, \\ 4x-8 & \text{if } x > 3 \end{cases}$ at $x = 3$.

3. $f(x) = \dfrac{x+1}{(x-3)^2}$ at $x = 3$.

4. $f(x) = \begin{cases} x\sin(\frac{1}{x}) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$ at $x = 0$.

5. $f(x) = \begin{cases} \frac{\sin(x)}{x} & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$ at $x = 0$.

6. $f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$ at $x = 0$.

7. $f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$ at $x = \pi$.

3 Continuity on an Interval

Definition 3.1 (Continuity on an interval)**.** (a) If f is defined on an open interval (a,b), then we say f is *continuous on* (a,b) if f is continuous at every point of (a,b).

(b) If f is defined on a closed interval $[a,b]$, then we say f is continuous on $[a,b]$ if f is continuous on (a,b), $\lim_{x\to a^+} f(x) = f(a)$, and $\lim_{x\to b^-} f(x) = f(b)$. We define continuity on a half open interval or an infinite interval similarly.

(c) More generally, we say f *is continuous* if f is continuous at every point in its domain.

Example 3.2. We have previously proved that the following functions are all continuous (i.e., continuous at every point in their domain):

- polynomials
- rational functions
- trig functions
- radical functions
- exponential functions
- the absolute value function

Using list above together with the Limit Laws, we can prove many other functions are continuous.

Theorem 3.3. *If f and g are continuous at a and c is a constant, then the following functions are continuous at a:*

(a) $f \pm g$

(b) cf

(c) fg

(d) $\frac{f}{g}$ *if* $g(a) \neq 0$.

Proof. Since f and g are continuous at a, $\lim_{x\to a} f(x) = f(a)$ and $\lim_{x\to a} g(x) = g(a)$. It follows that

$$\begin{aligned}
\lim_{x\to a}(f+g)(x) &= \lim_{x\to a}(f(x)+g(x)) \qquad \text{(by def of } f+g\text{)} \\
&= \lim_{x\to a} f(x) + \lim_{x\to a} g(x) \qquad \text{(by the Sum Law)} \\
&= f(a)+g(a) \qquad \text{(since } f,g \text{ are continuous at } a\text{)} \\
&= (f+g)(a).
\end{aligned}$$

The proofs of the remaining rules are similar and are left as an exercise. □

Exercise 3.4. Complete the proof of Theorem 3.3.

Exercise 3.5. Use Theorem 3.3 to determine where the function $\dfrac{x^3\sqrt{x-11}}{2x-9}$ is continuous.

Exercise 3.6. Find the value (or values) of k that makes each function below continuous over the given interval.

(a) $f(x) = \begin{cases} 3x+2, & x < k, \\ 2x-3, & k \le x \le 8 \end{cases}$

(b) $f(x) = \begin{cases} e^{kx}, & 0 \le x < 4, \\ x+3, & 4 \le x \le 8 \end{cases}$